

Title: Somatotopic Organization of Brainstem Analgesic Circuitry

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Abstract: The lateral periaqueductal gray (LPAG) evokes somatotopically appropriate defensive behaviours, including an analgesia that allows the animal to escape or fight unimpeded. Whether the LPAG and its descending targets are also able to drive somatotopically specific analgesic responses is not known. In ninety-three participants, we assessed LPAG ultra-high field functional magnetic resonance imaging signals during a placebo analgesia paradigm performed at different body locations. We found that analgesic responses are somatotopically organized in the LPAG, and its descending outputs to the rostral ventromedial medulla. These data show that the PAG can regulate analgesic responses in a highly spatially localised manner and thus has the ability to mediate body site-selective control over pain.

Main Text:

A paramount need for all living organisms is to preserve bodily integrity and prevent harm. Organisms faced with pain and other signs of danger to bodily integrity must make rapid choices in a context-sensitive manner, including the classically described ‘fight or flight’ decision (1). Experimental animal investigations have shown that these active defensive behaviours are integrated by neurons in the periaqueductal gray (PAG), an integrative midbrain structure that plays key roles in producing coordinated responses to threat. The PAG has a columnar structure, with dorsolateral, ventrolateral (vIPAG), and lateral PAG (lIPAG) playing distinct roles in different behaviors. Where the vIPAG is central for so-called ‘passive coping’ responses to threat, the lIPAG mediates active ‘fight’ or ‘flight’ responses (2, 3). lIPAG neurons receive direct inputs from cells in the dorsal horn and spinal trigeminal nucleus (SpV), providing direct access to nociceptive systems conveying pain-related information from the body and face, respectively. These projections are organized in a crude somatotopic fashion, that is SpV neurons project to the rostral lIPAG, whereas dorsal horn neurons project to the caudal lIPAG (4, 5). This somatotopy is linked directly to behavioural state changes, with rostral lIPAG stimulation evoking confrontation/fight behaviours, and caudal lIPAG stimulation evoking escape/flight behaviours (3). This organizational lIPAG scheme is conserved across species and appears to also be a feature of human PAG organisation, with a recent ultra-high field functional magnetic resonance imaging (fMRI) study reporting rostro-caudal lIPAG activation patterns during noxious stimulation to the face compared with the body (6). Although the lIPAG is modulated by inputs from higher brain regions (7), it can produce defensive behaviours without descending input (8). The somatotopically organised projections to, and from, the lIPAG drive spatially appropriate behavioural responses in an essentially reflexive manner, which confers a potentially life-saving behavioural importance to these circuits.

A key component for the successful execution of active defensive behaviours is a profound analgesia, which allows the behavioural response to be expressed unimpeded following an initial painful event (9). We know that spatially restricted analgesia is possible since expectancy of pain relief in one body region can evoke pain reduction in that region but simultaneously have no pain-relieving effects at other body sites (10). Whether the lIPAG also drives spatially appropriate, and restricted analgesic responses remains unknown. One way to explore the organization of lIPAG analgesic responses is using a placebo analgesia paradigm. It is well known that when one expects pain relief, an inert treatment can produce a profound reduction in perceived pain intensity, i.e. placebo analgesia. Placebo analgesia is mediated by forebrain regions that recruit brainstem circuits to inhibit noxious inputs at the level of the dorsal horn or SpV (7, 11). While it has been a long-held view that the brainstem circuit recruited during placebo analgesia depends specifically on neurons in ventrolateral column of the PAG, a recent fMRI study found that the lIPAG column is primarily responsible (12). The core difference between these functional PAG columns is the opiate-driven nature of analgesia elicited by the ventrolateral PAG, which when stimulated or lesioned produces a whole-body and long-lasting analgesia (13, 14). Conversely, lIPAG stimulation produces a spatially restricted and short-lasting analgesic effect. Given the established involvement of the lIPAG in placebo analgesia, it follows that relieved pain through effectively producing this phenomenon should also be spatially restricted to the body site where response conditioning or expectancy-modulation takes place.

Since placebo analgesia results from activity changes in the IPAG, which has a clear somatotopic organisation, it is possible that the circuitry underpinning placebo analgesia shares this somatotopy. Specifically, that placebo analgesia induced on the face and the body are driven by the rostral and caudal IPAG, respectively. This organizational principle may also be conserved in the major descending output targets of the IPAG that drive analgesic responses, for example, the rostral ventromedial medulla (RVM). Thus, specific parts of the RVM may preferentially contact and modulate incoming noxious inputs from the head, whereas other RVM regions contact the dorsal horn and modulate noxious inputs from the body. Such an organization would substantially advance our understanding of how the brain controls pain and would reveal a level of regional analgesic control previously unappreciated. We hypothesise that analgesia in the face and body will be paralleled by distinct somatotopic patterns of activity in the IPAG - RVM circuitry.

Behavioural placebo analgesia responses can be conditioned on separate body sites within the scanner environment. By deceptively applying different intensity thermal stimuli onto sites in the face, arm and/or leg, we conditioned healthy participants (face n=45, arm n=46, leg=41) to believe that a placebo cream (a sham “lidocaine”) was acting to reduce their pain relative to an adjacent control cream (“vaseline”). In a subsequent session, whilst collecting ultra-high-field (7-Tesla) fMRI, we applied identical intensity stimuli to both cream-sites (“vaseline”/control; “lidocaine”/placebo) and recorded the subjective pain intensity continuously using a visual analogue scale (VAS) extending from 0 (no pain) to 100 (worst pain imaginable) (Fig 1A). At each of the three body sites, a main effect of placebo was identified, such that across our entire cohort of participants pain was significantly reduced on the “lidocaine” placebo relative to Vaseline control side (paired t-test, $p < 0.05$). Subsequent effect size analyses revealed a moderate effect size for face placebo ($d=0.55$), small to moderate for arm ($d=0.39$), and a moderate to large effect for leg ($d=0.67$) (Supplementary Fig 1). However, this finding includes a distribution of significant placebo analgesia responders as well as non-responders. As such, we then classified participants as either responders or non-responders for each stimulus site and conducted group-level analyses using SPM12 to explore changes in signal intensity associated with evoking these analgesic responses. Despite all participants expecting reduced pain on the placebo-treated site (Fig 1D; Table 1), only 49%, 48% and 61% of participants demonstrated significant pain reductions when identical stimuli were applied to both sites on either the face, arm, or leg, respectively (Fig 1C; Table 1). Two way ANOVA with both group (responder, non-responder) and skin site (face, arm, leg) entered as separate factors revealed that these proportions were not affected by either age ($F[1,126]=1.06$, $p=0.31$) or sex distribution ($F[1,126]=2.71$, $p=0.10$), or by thermode temperature ($F[1,126]=0.27$, $p=0.61$) (Table 2, Supplementary Table 1).

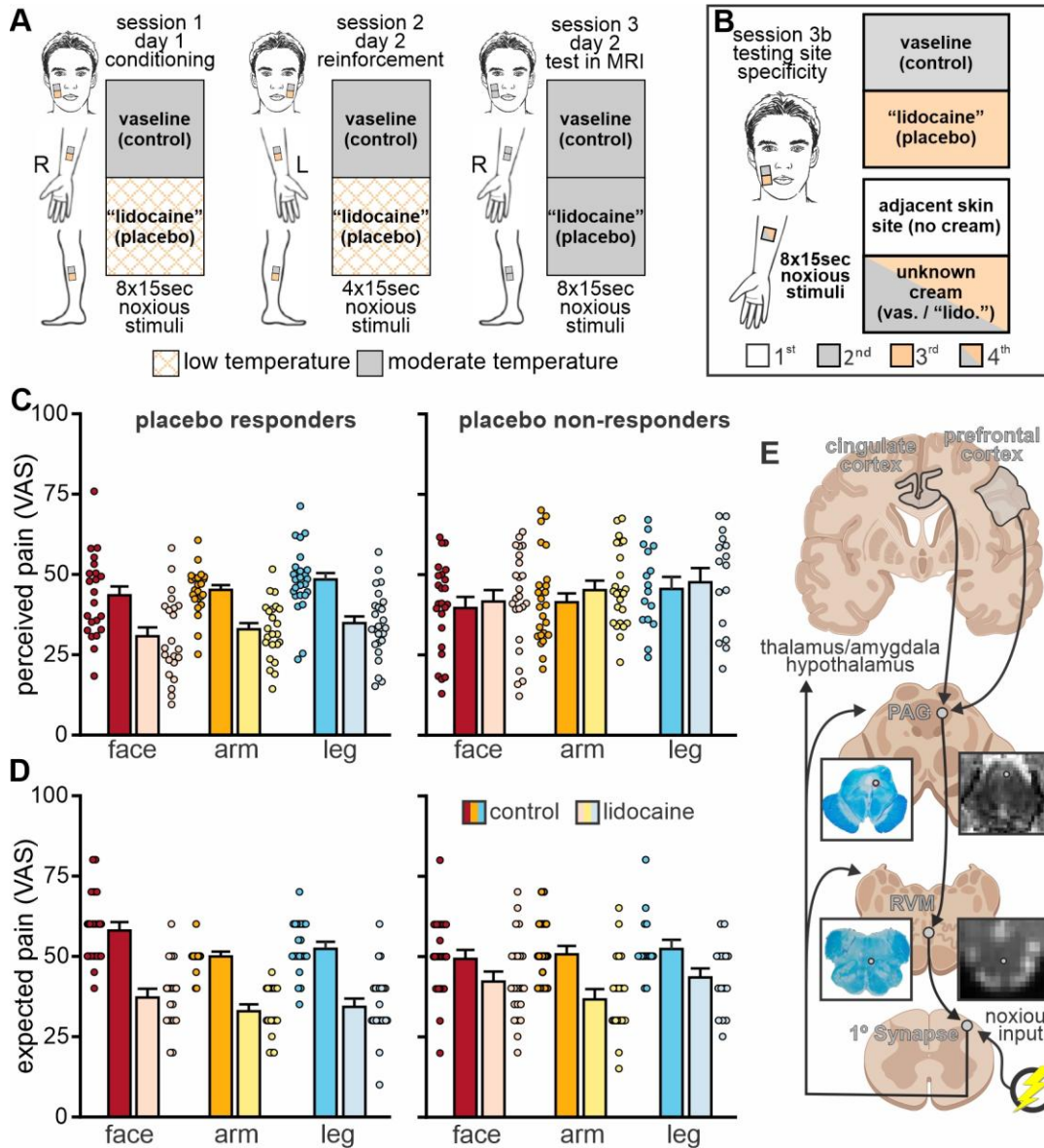


Fig 1: Experimental protocol and placebo-related pain changes. A) Placebo response paradigm. Conditioning was performed by applying low intensity noxious stimuli to the “lidocaine” site and moderate intensity to the vaseline site; crucially, during this phase participants deceptively believed stimuli of moderate intensity were being applied to both sites. On the following day, a reinforcement phase was conducted using the low and moderate temperatures on the opposite side of the body. Following a washout period, two independent functional magnetic resonance imaging (fMRI) series were collected where identical moderate intensity noxious stimuli to the control vaseline (scan 1), and placebo “lidocaine” cream (scan 2) sites were applied sequentially. During these two series, participants rated their expected and perceived pain on an MR-compatible visual analogue scale (0 = no pain, 100 = worst pain imaginable). R: right, L: left. **B) Unknown cream induction to determine systemic effects.** In those where response conditioning was conducted on the face, a third unknown cream was applied to the forearm without participants being informed or able to see which of the two creams this was. Immediately prior to testing evoked placebo analgesia responses to the face in these participants (n=23), an adjacent

untreated skin site of the forearm received repeated moderate noxious intensity stimuli, and following control and placebo “lidocaine” stimulation of the face, the unknown cream on the forearm also received these moderate noxious stimuli in the sequential order of: 1) arm no cream, 2) face control cream, 3) face placebo cream, 4) arm unknown cream. **C) Perceived pain intensities during control and placebo stimulations.** Combined bar and scatter plot displaying the distribution of perceived pain scores during control and placebo cream stimulation. Bar plot cut-offs indicate mean perceived pain intensity ($\pm$ SEM) with adjacent coloured circles indicating the range and spread of individual points comprising this bar, respectively, across the face, arm and leg in placebo responder and non-responder groups. **D) Expected pain intensities during control and placebo stimulations.** Combined bar and scatter plot displaying the distribution of expected pain intensity collected immediately prior to control and placebo cream stimulation. Bar plot cut-offs indicate mean expected pain intensity ($\pm$ SEM) with adjacent coloured circles indicating the range and spread of individual points comprising this bar, respectively, across the face, arm and leg in placebo responder and non-responder groups. **E) Schematic representation of likely cortical sites contacting descending pain modulatory circuitry during placebo evoked pain inhibition.** Long range projection neurons largely arising from the anterior cingulate cortex reach the midbrain PAG, with input from the dorsolateral prefrontal cortices both directly and via relay with the cingulate. The PAG then projects to the spinal cord dorsal horn where incoming noxious stimuli first synapse both directly yet sparsely, and predominantly via relay with the RVM. This same descending projection pattern from the PAG and RVM are found in the spinal trigeminal nucleus, where orofacial noxious input first synapses.

Table 1: Mean ($\pm$ SEM) pain ratings from responders and non-responders across face, arm and leg sites prior to (expected) and during (perceived) stimulation of the control and lidocaine (placebo) cream sites. The grey shaded boxes indicate where pain (expected or perceived) was significantly lower during the placebo compared with the control scans. Note that for the non-responders there was a significant increase in perceived pain during the placebo compared with the control scan.

	responders			non-responders		
	face (n=22)	arm (n=22)	leg (n=25)	face (n=23)	arm (n=24)	leg (n=16)
<i>expected pain ratings</i>						
control	58.4 $\pm$ 2.2	49.3 $\pm$ 0.8	52.9 $\pm$ 1.6	49.8 $\pm$ 2.4	51.2 $\pm$ 1.9	52.8 $\pm$ 2.3
placebo	37.7 $\pm$ 2.4	33.5 $\pm$ 1.6	34.8 $\pm$ 2.0	40.4 $\pm$ 2.1	37.1 $\pm$ 2.6	43.4 $\pm$ 2.6
p-value	p<0.001	p<0.001	p<0.001	p<0.001	p<0.001	p<0.001
effect size	1.91	2.65	1.99	0.86	1.27	0.96
<i>perceived pain ratings</i>						
control	43.5 $\pm$ 2.6	45.2 $\pm$ 1.5	48.4 $\pm$ 2.0	40.3 $\pm$ 2.9	42.2 $\pm$ 2.8	46.3 $\pm$ 3.1
placebo	30.7 $\pm$ 2.6	33.0 $\pm$ 1.9	34.9 $\pm$ 2.1	42.4 $\pm$ 3.2	45.9 $\pm$ 2.4	48.4 $\pm$ 3.8
p-value	p<0.001	p<0.001	p<0.001	P=0.06	p=0.04	P=0.38
effect size	1.07	1.52	1.32	0.14	0.28	0.15

Table 2: Mean ($\pm$ SEM) age, sex distribution and thermode temperature for responder (R) and non-responder (NR) groups for face, arm, and leg placebo. * $p < 0.05$ R vs NR.

face		arm		leg	
R	NR	R	NR	R	NR
<i>age (mean$\pm$SEM years)</i>					
27.9 $\pm$ 1.6	28.5 $\pm$ 2.2	24.0 $\pm$ 0.7	24.6 $\pm$ 0.9	24.1 $\pm$ 0.7	25.6 $\pm$ 1.1
<i>sex distribution (male:female)</i>					
10:12	13:10	14:8	10:14	12:13	8:8
<i>thermode temperature (mean$\pm$SEM degrees celsius)</i>					
46.2 $\pm$ 0.2	46.3 $\pm$ 0.2	46.9 $\pm$ 0.2	46.8 $\pm$ 0.2	46.4 $\pm$ 0.1	46.2 $\pm$ 0.2

Response Conditioning on distinct body sites elicits a regionally specific placebo analgesia:

Within a sample of 23 participants undergoing face placebo, we applied a third cream to the right forearm during the MRI session immediately prior to commencing the two face placebo scans. This cream was described to be one of either the control Vaseline or “lidocaine” (placebo) – and whilst participants could feel the creams’ application, they could not observe the bottle nor label. Four series of repeated and identical moderate intensity noxious stimuli were applied to: series 1: an adjacent skin site of the forearm; series 2: the face control cream site; series 3: the face placebo “lidocaine” cream site; and series 4: the unknown forearm cream site (Fig 2A). In these participants, a main effect of pain change was observed between the two cream sites of the face ($n=23$), with this effect intensified following permutation testing and isolation of face placebo responders ($n=13$) (Figs 2B/2C, Table 3). Comparison of pain intensity ratings between noxious stimulation of the adjacent skin site of the forearm (series 1), and the unknown cream site (series 4) revealed no significant reduction in pain either at a group level or when isolating only those participants which showed a significant analgesic response on the face (Figs 2B/C). **Two way repeated measures ANOVA revealed an interaction effect between condition and skin site in specifically placebo responders ($F[1,12]=9.52$, $p=0.009$), however not across all twenty-three participants ($F[1,22]=2.50$, $p=0.13$).** Moreover, linear regression analysis revealed no significant interaction between overall pain change experienced on the face relative to the forearm sites ($R=0.25$, $p=0.23$) (Fig 2D), with a response ratio of 56% ($n=13$) and 17% ($n=4$) on the two known cream versus hidden cream sites of the face and forearm, respectively.

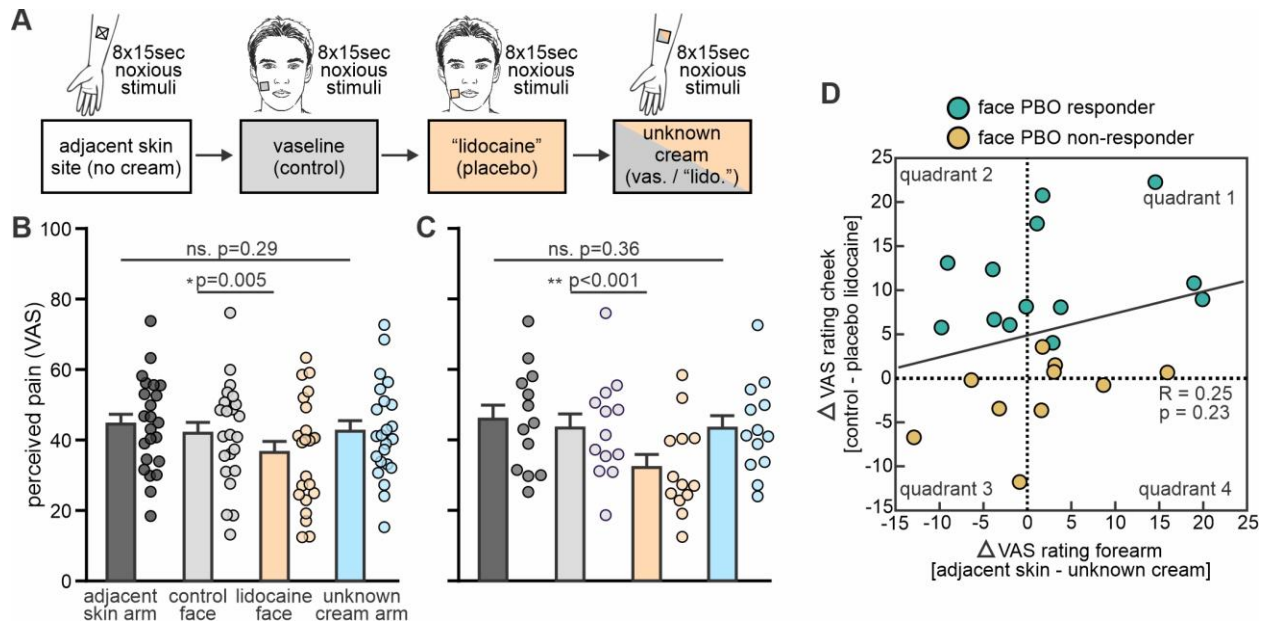


Fig 2: Response conditioning elicits a body-site specific placebo analgesia that is not expressed through subsequent application of an unknown cream distally. A) Methodology.

In participants where placebo analgesia was conditioned to the face, a baseline pain measurement on an untreated skin site of the forearm was conducted, before a single cream was applied adjacent to this site described as being either the Vaseline (control) or “lidocaine” (placebo) previously used throughout conditioning. Participants were aware of this cream’s application, however, could not view which cream was applied with vision blocked by the MRI scanner head coil. Following noxious stimulus application to both the control Vaseline and placebo “lidocaine” cream sites of the face, the unknown cream was removed from the forearm and repeated identical intensity noxious stimuli applied. If conditioning to the face induced a body-wide placebo response, a reduction in pain would be observed on the unknown cream site, however, if this analgesic response is site-specific, no change in pain should be observed between the untreated- and unknown cream sites of the forearm.

B) Main effect perceived pain intensity of arm and face. In each participant ($n=23$), overall mean pain intensity across the eight noxious stimuli applied to either the forearm or face on each cream site and adjacent skin site of the forearm were calculated. A significant reduction in perceived pain was observed between the two cream sites of the face, with no significant change in pain reported between the untreated skin site and unknown cream site of the forearm. **C) Face placebo responder cohort pain intensity of forearm and face.** This pattern of restricted analgesia to the body site where response conditioning was conducted was further intensified when comparing only those participants who, revealed through bootstrapped permutation testing, expressed a significant behavioural placebo analgesia on the face ($n=13$). **D) Perceived pain intensity changes of the face and arm show no significant linear relationship.**

Comparing each participants magnitude of pain intensity change between either the control Vaseline and placebo “lidocaine” cream sites of the face, or untreated skin or unknown cream sites of the forearm revealed that participants largely evoked either solely a significant placebo analgesia response on the face alone (Quadrants 1 and 2), or a placebo analgesia response on neither body location (Quadrant 3). Vas. = Vaseline; lido = lidocaine; Ns. = no statistical significance; PBO = placebo; VAS = visual analogue scale.

Table 3: Mean ($\pm$ SEM) pain ratings during stimulation of untreated skin of the forearm, control and lidocaine (placebo) creams of the face, and unknown cream of the forearm to assess regional specificity of placebo analgesia. Gray shaded boxes indicate significant decreases in perceived pain – note that pain was only reduced between face skin sites and not between forearm sites away from the site of response conditioning following expression of placebo analgesia. Linear regression further revealed no significant interaction between behavioural placebo analgesia elicited on the face versus perceived pain change on the forearm.

	perceived pain	p-value	t-value	effect size
main effect face sites (n=23)				
control	41.9±2.9	0.005	3.08	0.64
placebo	36.5±3.1			
main effect forearm sites (n23)				
untreated Skin	44.5±2.7	0.29	1.08	0.26
unknown Cream	42.5±2.8			
responder only face sites (n=13)				
control	43.4±3.9	<0.001	6.77	1.88
placebo	32.3±3.5			
responder only forearm Sites (n13)				
untreated skin	46.0±3.9	0.36	0.77	0.27
unknown cream	43.5±3.5			

Placebo analgesia somatotopically engages discrete nuclei of the midbrain periaqueductal gray and rostral ventromedial medulla: Consistent with our hypothesis, analysis of fMRI activation patterns in placebo responders compared with non-responders revealed a crude somatotopic organization within the PAG (Fig 3A, Table 4). That is, placebo analgesia on the face evoked signal intensity change differences in the rostral PAG, whereas placebo analgesia on the arm and leg evoked signal differences more caudally in the PAG (Z co-ordinates; face: -5, arm: -9.5; leg: -9.0). Consistent with this organization, linear regression analysis of signal intensity change differences (control versus “lidocaine” scans) with analgesic ability also revealed the same somatotopic pattern. Signal intensity changes were negatively correlated with analgesic ability and localised to the rostral PAG for the face (Z co-ordinate face: -4) and, again, more caudally for the body sites (Z co-ordinates; arm: -9.5; leg: -9.5) (Fig 3B, Table 4). Each of these six clusters survived small volume correction at a family wise error rate of $p < 0.05$. Additionally, single and between-group regression revealed these correlations were unique for each placebo site. That is, while signal change differences for each cluster derived from the placebo face analysis were correlated with analgesic ability, values extracted from this cluster for the arm or leg placebo scans were not significantly correlated. Moreover, comparison of each significant regression slope in one body site against the regression slope generated between extracted signal change values and analgesic ability in the alternate two body sites yielded significantly different correlation coefficients in each of the three PAG clusters (Table 5). For all three stimulus sites, placebo analgesia responders displayed signal intensity change decreases whereas non-responders displayed signal intensity change increases.

Table 4: Mean peak X, Y and Z Montreal Neurological Institute (MNI) co-ordinates and t-values in the ipsilateral midbrain periaqueductal gray matter (PAG) and rostral ventromedial medulla (RVM) during placebo analgesic responses applied to the face, arm and leg.

	MNI co-ordinate			t-value	responder beta-value (mean±SE M)	non-responder beta-value (mean±SEM)
	X	Y	Z			
<i>PAG negative correlation with placebo ability</i>						
Face	3.0	-27.5	-4.0	3.15	-	-
Arm	2.0	-34.5	-9.5	3.10	-	-
Leg	1.5	-30.5	-9.5	3.40	-	-
<i>PAG responder vs non-responder</i>						
face	2.5	-29.0	-5.0	3.59	-0.46±0.33	1.09±0.31
arm	2.5	-34.5	-9.5	2.85	-0.54±0.32	1.05±0.39
leg	2.0	-30.5	-9.0	2.64	-0.48±0.23	0.87±0.41
<i>PAG individual nadir signal change analysis (mean±SEM)</i>						
face	1.52±0.11	-31.32±0.47	-6.54±0.64	-	-	-
arm	1.78±0.14	-32.39±0.54	-9.05±0.65	-	-	-
leg	1.78±0.14	-32.48±0.45	-8.38±0.58	-	-	-
<i>RVM positive correlation with placebo ability</i>						
face	3.0	-35.5	-41.5	2.98	-	-
arm	1.5	-40.0	-46.5	3.34	-	-
leg	1.0	-36.5	-48.5	3.67	-	-

Table 5: Upper panel. Extracted regression coefficients and p-values between signal intensity change difference (placebo lidocaine – control scan) and perceived pain change from each significant cluster of the ipsilateral midbrain periaqueductal gray matter (PAG) and rostral ventromedial medulla (RVM) that correlated with placebo analgesia evoked on the face, arm, or leg. * Indicates significance, correlations between activation change and perceived pain change were only observed in the specific body site of placebo analgesia where each cluster emerged, and not with the other two body sites. Lower panel. Between-group comparison of signal intensity change difference and perceived pain change in significantly activated clusters of the PAG and RVM in either face, arm, or leg placebo. * indicates significance between regression slopes when beta-values were extracted from alternate body sites and compared to the body site which initially yielded significance in voxel-by-voxel analyses. Regression lines between groups were significant for both the PAG and RVM in all body sites with exception of the RVM cluster significantly correlated with leg placebo.

<i>Single Group Regression Coefficient</i>						
	face R-Value	p-value	arm R-value	p-value	leg R-value	p-value
face PAG	-0.44	0.002*	-0.10	0.51	-0.01	0.98
face RVM	0.46	0.001*	0.03	0.85	0.10	0.55
arm PAG	-0.02	0.88	-0.44	0.002*	0.08	0.58
arm RVM	0.14	0.36	0.47	0.001*	0.08	0.63
leg PAG	-0.06	0.66	-0.13	0.40	-0.49	0.001*
leg RVM	0.26	0.09	0.05	0.74	0.50	0.001*
<i>between group Regression Slope Comparison</i>						
	face x arm p-value		face x leg p-value		arm x leg p-value	
face PAG	0.04*		0.01*		0.54	
face RVM	0.02*		0.002*		0.71	
arm PAG	0.04*		0.91		0.008*	
arm RVM	0.03*		0.57		0.02*	
leg PAG	0.64		0.04*		0.04*	
leg RVM	0.18		0.91		0.01*	

Individual analysis of greatest (nadir) differences between control and placebo scans in responders again confirmed a rostro-caudal PAG organization (mean $\pm$ SEM Z-coordinate: face -6.54 ± 0.64 ; arm: -9.05 ± 0.65 ; leg: -8.38 ± 0.58 ; Fig 3B, Table 4). A one-way ANOVA revealed that there was a statistically significant difference in the mean ipsilateral Z co-ordinate between at least two body sites ($F(2, 69) = 4.44$, $p=0.01$). Tukey's HSD test found that the mean Z co-ordinate was significantly different between both the face and arm ($p=0.01$), and face and leg ($p=0.04$), but not arm and leg ($p=0.88$), revealing a somatotopic organisation between face and body placebo. Both responder/non-responder and correlation group analyses resulted in clusters located across different PAG columns. However, analysis of the X, Y, Z co-ordinates of nadir signal intensity change differences in responders, revealed the mean locations for the face, arm and leg were localised within the LPAG column when overlaid onto a mean brainstem fMRI image set created from all participants (Fig 3C).

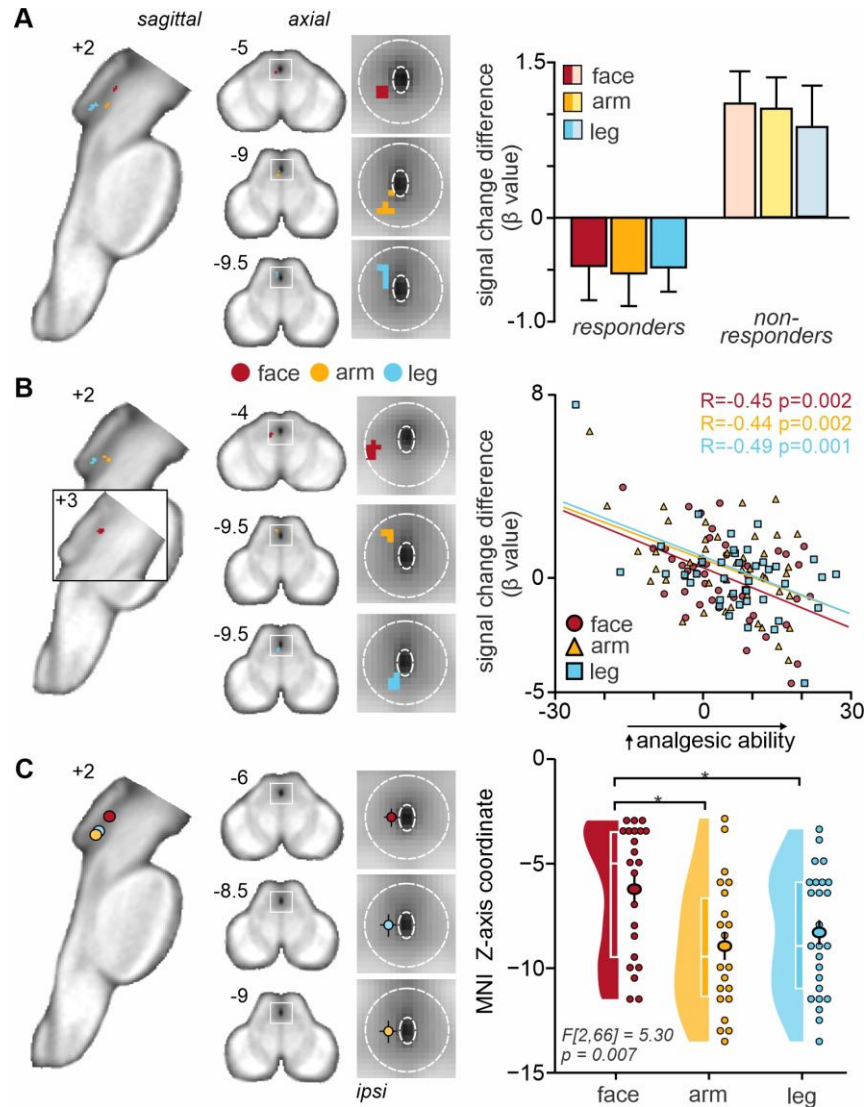


Fig 3: Somatotopic organization of the PAG during placebo analgesia. A) Responder versus non-responder. Analysis of noxious stimuli evoked signal intensity change differences (placebo lidocaine-control scan) revealed a somatotopic PAG organization with placebo analgesia on the face evoking signal intensity change differences (placebo-control) in the rostral PAG, and placebo analgesia on the arm and leg evoking signal change differences more caudally in the PAG ($p < 0.005$, uncorrected for multiple comparisons, cluster extent threshold = 5). To the right are plots of signal intensity change differences (β value change) for responders and non-responder for each site. **B) Signal change correlation with analgesic ability.** Analysis of noxious stimulus evoked signal intensity change differences correlated with analgesia strength also revealed a rostro-caudal, face-body, PAG organization ($p < 0.005$, uncorrected for multiple comparisons, cluster extent threshold = 5). To the right are plots of signal intensity change differences versus analgesia strength for each site. **C) Individual responder participant signal intensity change differences.** Analysis of individual signal intensity change differences in responders also revealed a rostro-caudal organization for face versus arm placebo. Furthermore, these peaks were located immediately lateral to the aqueduct within the boundary of the lateral PAG column. To the right

are raincloud plots of individual peaks for the face, arm and leg placebo with the mean and median shown by the circle and box plot, respectively. * $p < 0.05$.

Also consistent with our hypothesis we found that placebo analgesia preferentially activates distinct regions of the RVM depending on whether analgesia was elicited on the face or body. Although a responder versus non-responder analysis of the face, arm and leg did not reveal significant RVM signal intensity change differences, a correlation analyses revealed significant positive correlations for each placebo site in distinct RVM regions. That is, signal intensity changes were positively correlated with analgesic ability within different RVM locations during face, arm, and leg placebo scans, **with each of these clusters surviving small volume correction at a family wise error rate of $p < 0.05$** (Fig 4, Table 4). Face placebo was associated with signal intensity correlations more rostral than that evoked by arm and leg analgesia responses Z coordinates; face: -41.5, arm: -46.5; leg: -48.5), such that similar somatotopic activity change was conserved between the PAG and RVM. As in the PAG, correlations between RVM signal intensity change and analgesic ability were unique to each site, with significant positive interactions only observed in the specific body site where each of the three clusters emerged. **Additionally, with exception of the comparison between the shin RVM cluster in the face placebo condition, significantly different between-group correlation coefficients were observed in each of the three RVM clusters (Table 5).**

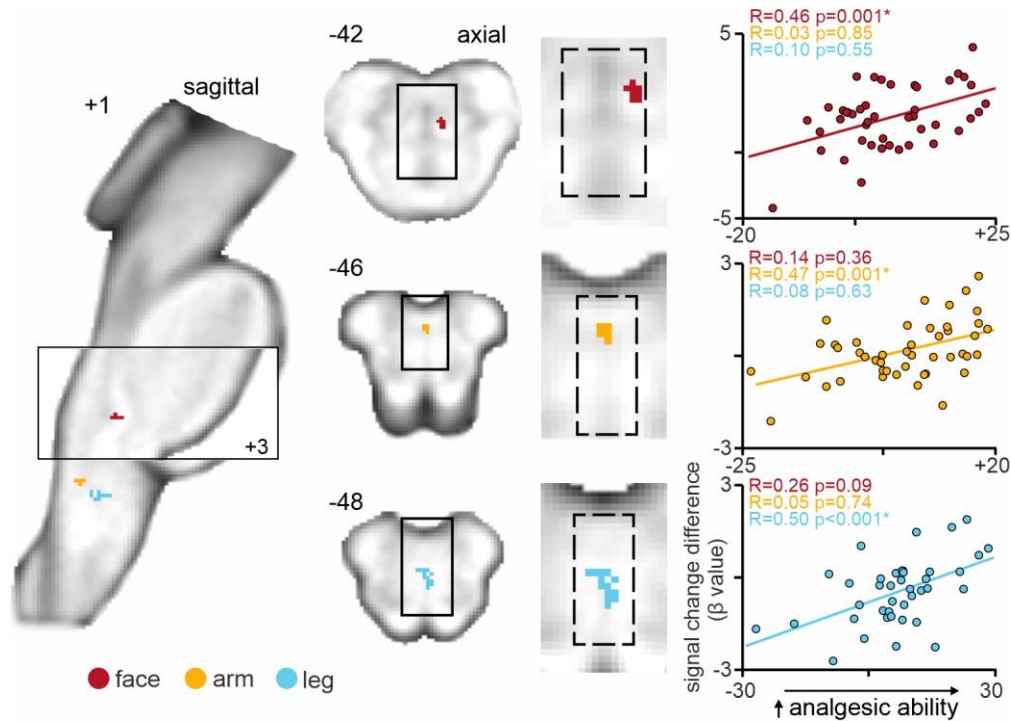


Fig 4: Organization of the RVM during placebo analgesia. Signal intensity change difference correlated with analgesic ability. Analysis of noxious stimuli evoked signal intensity change differences (placebo lidocaine-control scan) revealed a discrete RVM organization when placebo analgesia is elicited on the face, arm and leg. Placebo analgesia on the face was associated with a signal correlation within a more caudal aspect of the RVM when compared with placebo analgesia evoked on the arm and leg. To the right are plots of signal intensity change differences (β value change) correlated against placebo analgesia ability for each site. Note that for each site, the cluster

identified was uniquely correlated with analgesia strength. That is, the cluster identified during the face placebo voxel-by-voxel analysis was significantly correlated only for the face placebo scan and not for the arm or leg placebo scans; the same applies for arm and leg placebo clusters.

Engagement of contralateral midbrain during pain processing informs location and intensity of ipsilateral midbrain contact during placebo analgesia: Individual analysis of maximum signal intensity change values within the contralateral PAG during the control scan alone revealed a similar somatotopic pattern of PAG activation during pain processing. That is, maximum activated voxels during noxious stimulation of the face were represented more rostral to both the arm and leg (mean $\pm$ SEM Z-coordinate: face: -6.54 ± 0.64 , arm: -8.32 ± 0.49 , leg: -8.56 ± 0.58). In each responder cohort across the three body sites ($n=69$), linear regression was conducted comparing the maximally activated Z co-ordinate in the contralateral PAG with the nadir point ipsilaterally, revealing a significant linear association ($R=0.32$, $p=0.007$). That is, placebo responders on the face activated a similarly more rostral component of the PAG during pain processing alone as in when expressing placebo analgesia. Likewise, placebo analgesia elicited to the upper and lower limbs was associated with more caudal activity in the contralateral PAG during pain processing, and the greatest change in activity in the ipsilateral PAG during placebo analgesia expression (Fig 5A). Importantly, the average point of maximum activity during pain, similar to the nadir point during placebo analgesia, was isolated to the lateral PAG column. Moreover, the signal intensity within each individual's maximally activated PAG voxel in the contralateral side was also compared through linear regression to the change in activity in the ipsilateral side, revealing a significant association was also present in the magnitude of activation change across the two sides of the lateral PAG ($R=0.45$, $p<0.001$) (Fig 5B).

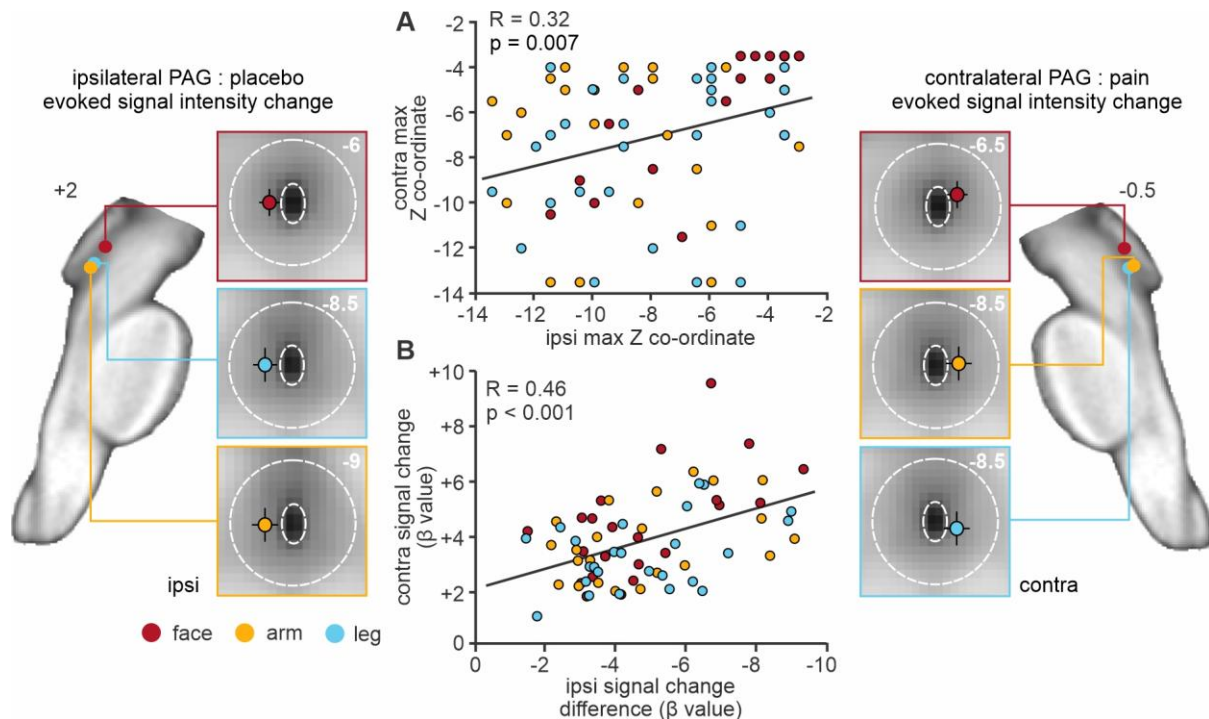


Fig 5: Somatotopic organization of the midbrain periaqueductal gray matter during both pain processing and placebo analgesic responses. A) Ipsilateral versus contralateral localization of placebo-evoked and pain-evoked signal changes in the midbrain. In each participant that demonstrated a significant placebo analgesia response on either the face, arm, or leg (n=69), the Z co-ordinate of the ipsilateral periaqueductal gray (PAG) which showed the largest stimuli evoked signal intensity change differences (placebo lidocaine-control scan) and the Z co-ordinate of the contralateral PAG which showed the largest stimuli evoked signal intensity change (control scan only) were extracted and compared through linear regression. This revealed a significant relationship such that both nadir and maximum points of the PAG ipsilateral and contralateral respectively were largely more rostral when conditioned placebo analgesia was evoked on the face, and more caudal when evoked on the arm and leg. **B) Ipsilateral versus contralateral beta-value magnitude in placebo expression and pain processing of the midbrain PAG.** Similarly, a significant positive relationship was observed between beta-values encoding the largest signal intensity change difference and signal intensity change in the ipsilateral and contralateral PAG, respectively. That is, those participants who evoked a significant placebo analgesia response on either the face, arm, or leg who had the largest increase in signal intensity in the contralateral PAG during pain processing (control scan only) also demonstrated the largest change in activation during placebo expression (placebo lidocaine-control scan).

Discussion: Our data reveals that brainstem circuits involving the PAG and RVM, activated by placebo analgesia, have a clear somatotopic organisation. We show that placebo analgesia elicited on the face, arm, and leg is associated with altered activity at specific locations of the PAG and RVM, supporting that the brainstem has the ability to selectively modulate incoming noxious inputs entering from a specific part of the body/head. Moreover, our data demonstrate that this specific placebo analgesia is spatially specific, with pain relief being produced only on the skin site where response conditioning and takes place, and not at distal body sites, suggesting systemic effects of pain relief are not involved. That is, by introducing a third hidden cream to a distal body site and stimulating this site after placebo analgesia expression, we demonstrate no association between relief in pain on the site of response conditioning versus this distal site. This specificity is consistent with the idea that during active defensive behaviours, the body regions in which noxious inputs occur, drive not only spatially appropriate behavioural responses, but that these responses are coupled with a somatotopically appropriate analgesia. Our findings extend those recently demonstrated by Mehnert and colleagues – who defined a highly granular dermatome-specific somatotopy of the PAG in orofacial noxious inputs (15) – by resolving how the upper and lower limbs are also represented in somatotopic organization along the rostro caudal extent of the PAG. Moreover, in a design inspired by (16) and through application of a hidden cream distally to the body site where placebo conditioning takes place, we reveal that the placebo analgesia we evoke is site-specific and is not accompanied by a body-wide downregulation of perceived pain. It has been proposed that at the level of the spinal cord, stimulation of large diameter non-noxious sensory afferents can interfere with incoming small-diameter afferent noxious inputs to produce a regionally specific analgesia, i.e. the pain-gate theory (17). Our data are the first to show that circuits within the human brain are also organized such that incoming noxious inputs can be modulated in a regionally-specific manner. Indeed, although no study has described a somatotopic organization of IPAG evoked analgesia, over 30 years ago it was reported in rodents that morphine injections into the lateral versus ventrolateral PAG evoke spatially restricted and widespread analgesia, respectively (18) and that dorsoventral PAG electrical stimulation may evoke analgesia in different parts of the body (19). With the precise organization of IPAG evoke analgesia now

resolved in humans, these findings fundamentally advance our understanding as to the regional specificity of analgesic pathways, providing evidence that the brain contains spatially separate circuits that can produce analgesia in specific body regions. These data also raise the prospect that altered function in these anatomically-specific brainstem analgesic circuits may underlie the presence of regionally restricted persistent pain.

Despite all participants expecting the placebo “lidocaine” cream to modulate pain intensity, in only a proportion of individuals was a significant reduction in pain intensity reported. This variability is consistent with reports that pain modulation relies on concordance between expectations and experience of prior painful events (20, 21). Despite this variability, we found that similar numbers of individuals displayed an analgesic response during placebo applied to the face (49%) and arm (48%), with a higher response ratio when evoking placebo analgesia on the leg (61%). Overall, these proportions of significant responses are similar to those reported previously (22), indicating that body site does not impact placebo analgesia rates. We also present contrary evidence to the accepted hypotheses that placebo analgesia is opioid-dependent (23-25) and by extension, is mediated by opiate-receptor binding within the ventrolateral PAG (26, 27). Rather, we identified consistently across all body sites that it is the IPAG which was responsible for placebo analgesia expression. Notably, the IPAG column is unresponsive to morphine microinjection and, when stimulated, drive antinociceptive behaviours without the influence of endogenous opiates (13, 28). Although group level analyses resulted in variability with respect to the columnar distribution of PAG signal changes for placebo on the face, arm and leg - similar to our previous study exploring PAG organization during noxious stimuli (6) - individual level analyses revealed a consistent peak signal change in the ipsilateral IPAG column. Notably, our findings are restricted to the likely neurotransmitter systems acting within the *brainstem* during placebo analgesia expression, and higher brain regions involved in expectation, such as the anterior cingulate cortex and amygdala, have been shown to be recruited during placebo analgesia (29-31). These higher-order areas activity can be directly modulated, and their output projection patterns altered by opioid microinjections, and both project strongly to the IPAG (32, 33). Therefore, our investigation does not dismiss that opioid activity acting on these sites could provide a key input in driving IPAG-mediated pain modulation.

The patterns of placebo-induced IPAG signal changes we define are consistent with the projection of afferent inputs from the dorsal horn and SpV (4, 5), as well as patterns of IPAG activation following noxious stimuli (34). The caudal IPAG: 1) receives projections from the contralateral spinal dorsal horn, 2) is activated by noxious stimulation of the contralateral forelimb and hindlimb (6), and 3) shows signal changes following placebo analgesia responses in the leg and arm. This same pattern exists in the rostral IPAG for stimuli from the face (6). The patterns of placebo-induced IPAG signal changes we define are consistent with the observations that the columns on each side of the PAG are bilaterally interconnected (35) which allows the PAG to generate bilateral output signals resulting in whole body coordinated responses. Contralateral noxious inputs can therefore drive ipsilateral output signals (ie., placebo analgesia), shaped by strong top-down descending inputs from frontal cortical regions with specific columnar targets. Our data suggest that such connections can provide somatotopically-directed analgesia by selective engagement of discrete components of the descending modulatory systems in response to somatotopically specific pain.

In addition to describing somatotopic organization in the IPAG, we also found that the RVM displays regionally specific signal intensity changes during face, arm, and leg placebo analgesia. It is well-established from experimental animal investigations that the IPAG modulates incoming noxious information via projections to the RVM, which then project directly to the SpV and dorsal horn (27, 36-39). The RVM contains both “off” and “on” neurons that strongly inhibit and facilitate neurotransmission at the primary nociceptive synapse, respectively (40-44). We found opposing relationships with greater analgesic ability associated with reduced PAG signal changes and increased RVM signal changes. These findings suggest that during placebo analgesia, reduced IPAG synaptic activity evokes an overall increase in RVM “off” compared to “on” cell firing, which, in turn, results in increased inhibition of incoming nociceptive drive at the dorsal horn or SpV.

Our results reveal that RVM signal changes during face, arm and leg placebo are largely discrete and spatially separated, consistent with the idea that the RVM contains groups of neurons that preferentially target specific parts of the dorsal horn or the SpV. While evidence for a somatotopic organization within the RVM is novel, earlier pre-clinical studies have shown a degree of anatomical specificity within the RVM. For example, anterograde tracing has revealed that midline neurons in the RVM (raphe magnus) project preferentially to deep laminae (V-VI), whereas more laterally placed RVM neurons (lateral paragigantocellular neurons) project to the superficial laminae (I-II) of the dorsal horn (45). Moreover, it has been reported that RVM projections to spinal dorsal horn are predominantly GABAergic and are more likely to contact neuronal soma, whereas projections to SpV have only modest numbers of GABAergic cells which preferentially target dendrites (46). We know of no preclinical study that has specifically addressed the question of somatotopic organization of the RVM, our human data shows that, like the IPAG, the RVM is activated in a regionally specific, rostro-caudal somatotopic manner during placebo-mediated analgesic responses.

It is important to note some limitations. Firstly, it was not possible to fully counterbalance the ordering of stimuli and as such our order was fixed to stimulate the control cream site always first and placebo lidocaine second since the experimental design required a pairing of modulated with non-modulated responses. This ordering effect may have increased the likelihood of habituation interfering with the effect of placebo analgesia expression. However, creams were placed in non-overlapping skin areas, and each series was separated by five minutes, which was the time taken to replace the thermode between the two cream sites after each functional series, reducing the likelihood of peripheral fatigue or carry-over habituation (47, 48). Secondly, conditioning-based pain modulation paradigms are prone to response bias (49). To counteract this bias, each participant rated their pain continuously during the scan instead of afterwards, which would have induced an additional cognitive aspect to the protocol. Since all participants performed pain ratings in the same manner, it is unlikely that this approach would have a significant effect on the results. Third, our initial statistical thresholding was relatively liberal for second-level contrast images, $p < 0.005$ uncorrected for multiple comparisons with a cluster extent threshold of 5 contiguous voxels. However, our results are limited solely to the PAG and RVM, two regions previously shown to be critical for placebo analgesia. Thus, signal changes in these regions are consistent with our hypothesis on these two specific regions and align with decades of preclinical studies

describing a role for these areas in pain modulation. Additionally, we utilized both dichotomization of participants as responder and non-responder cohorts as well as correlational analyses to generate our behavioural and imaging findings. These two methods are debated, and to our knowledge no clear consensus exists whether a dichotomization or continuous approach should be taken to placebo analgesia responses in humans. Importantly, the inclusion of both results sets bolster an identical finding, suggesting that aspects of PAG and RVM circuitry is involved in endogenous pain modulation through placebo intervention regardless of whether a dichotomization or regression approach is employed. Finally, our two sample results indicate that in placebo non-responders LPAG signal increases in a similar magnitude to the decreases observed in the placebo responder groups. Given these participants' report similar pain between the two scans, it is curious their overall magnitude of signal intensity change is similar to those that between the two scans report a large difference in pain. Notably, despite the *magnitude* of these signal changes being similar, their *directionality* is in opposition. The PAG is a region known to hold several complex roles not only in activating the descending analgesic system, but also in driving error-prediction and updating expectancies toward external environmental factors (50, 51). We suggest that despite showing no difference in reported pain, placebo non-responders are undergoing large behavioral changes between functional series, being forced to update their expectancies, and recruiting the PAG to drive this process, directly opposing the action of placebo responders where expectancies are met and perceived pain reduced, through the action of top-down circuits on discrete body-specific locations of the PAG.

Using ultra high-field fMRI we have shown a somatotopic organization within the PAG and RVM following placebo analgesia evoked on the face and body. A future understanding of whether this analgesia is expressed maximally at the site receiving response conditioning and reduced if the same conditioned cream is subsequently placed on increasingly distal body sites would be important for resolving how this stimulus-response association is established and maintained throughout conditioning, reinforcement, and testing phases. Ultimately, these findings provide evidence that the brain is capable of modulating incoming noxious information in a regionally-specific manner which may allow for the rapid coordination of the most appropriate defensive behaviors.

Materials and Methods:

Ethics: All experimental procedures were approved by the University of Sydney Human Research Ethics Committee (Project number 2019/037) and were consistent with the Declaration of Helsinki. Written informed consent was obtained from participants at the commencement of the study. Participants were provided with an emergency buzzer while inside the scanner so that they could stop the experiment at any time. At the conclusion of testing, participants were informed verbally, and through a written statement, of the necessary deception and true methodology of the experiment and were invited to seek clarification of what they had just experienced.

Participants: Ninety-three participants were recruited for the study (47 male, 46 female; mean $\pm$ SEM age, 26.1 $\pm$ 0.7 years; range 19 - 73 years). Inclusion criteria consisted of being over 18 years of age, able to provide written and informed consent, and no current or past diagnosis of any chronic pain condition or pain lasting longer than three months. Exclusion criteria consisted of the

presence of any standard MRI-contraindications (current pregnancy, ferrous implants, etc.), or current or past chronic pain diagnosis. In order to evaluate the necessary number of participants required for this study, an *a priori* power analysis was performed using results from a previous imaging study investigating cortico-brainstem connectivity during placebo analgesia (30). This revealed a within-subjects sample size of 40 at each body site would be necessary to detect similar effect sizes with 95% power ($d = 0.59$, $\alpha = 0.05$, power = 0.95). Before each day of the study, participants completed a data sheet recording current medication(s), and any alcohol or caffeine ingested in the 24 hours prior to testing, to ensure no significant changes in consumption of these substances were present between the two successive days where experiments took place. Of our 93 total participants, 19 were on medication at the time of conditioning and scanning: four on antihistamines, seven on oral contraceptive medications, two on antiretrovirals, four on antidepressants, one on a course of antibiotics, and one using a topical retinoid cream.

Experimental Design: The study included three sessions occurring on two successive days: a conditioning session on day 1, and a reinforcement and MRI scanning session on day 2 (Fig 1A). Participants enrolled to one of four slots which were listed with the following arrival times for each of the two days: 1) 9am / 9am; 2) 10:30am / 11am; 3) 12:00pm / 1:00pm; or 4) 1:30pm / 3:00pm - such that the time of day was kept consistent between experimental sessions. Throughout the study, noxious stimuli were administered to one of three places: the left and right cheek of the face, volar surface of the forearm on the arm, or on the shin of the lower leg. Brief noxious stimuli were delivered using a 3x3cm MR-compatible Peltier element thermode, which delivered a heat stimulus at a pre-programmed temperature via a Thermal Sensory Analyzer (TSA-II) (Medoc LTD Advanced Medical Systems, Rimat Yishai, Israel). Each stimulus lasted 15 seconds, including a ramp-up period (four degrees per second), a plateau period at a noxious temperature and a ramp-down period (four degrees per second). Each stimulus was separated by a 15 second inter-stimulus-interval (ISI) at a non-painful baseline temperature of 32°C. Throughout conditioning, participants rated their pain on-line using a horizontal 10 cm visual analogue scale (VAS) ranging between 0 and 100, where 0 was described as “no pain” and 100 as “the worst pain imaginable”. During scanning, participants used an MR-compatible button box to continuously report their pain perception. The VAS scale was shown on a reflected digital screen at the end of the magnet bore, and participants controlled the position of a slider to report their pain continuously by holding the left (moved slider towards zero) or right (moved slider towards ten) buttons with their left middle and index finger.

Placebo analgesia paradigms were applied on either the arm only ($n=27$), arm and leg ($n=21$), leg and face ($n=22$), or face alone ($n=23$). In participants where two sites were tested, the order was counterbalanced to limit any effects of ordering or sensitivity. In two participants in which the face and leg were tested, and two participants where the arm only was tested, significant motion artefacts occurred during leg and arm stimuli, respectively, and thus these datasets were removed from final analyses.

Conditioning: Session 1 was conducted outside the MRI and consisted of two rounds of a conditioning protocol. Participants were first informed both verbally and via a written statement that the study was designed to investigate the modulatory effects of a topical anaesthetic containing lidocaine, which had been shown to provide pain relief in some individuals. A second control

cream was stated to be purely Vaseline and was stated as being necessary to evaluate typical pain responses. In reality, both creams contained Vaseline, and differed only in colour and their described properties. Individual low and moderate pain responses were calculated by applying a series of ten stimuli in randomized order to the right face, arm, or leg at 0.5-degree intervals from 44-48.5 degrees Celsius, and asking participants to rate their perceived pain during the plateau period of each stimulus. Participants were informed that we were only recording a temperature which elicited a moderate subjective pain response (40–50 VAS rating), and that this temperature would be used throughout the remainder of the experiment. However, using the ratings provided during this process, we recorded two different temperatures: one which was rated between 20-30 on the VAS (low temperature); and one which was rated between 40-50 (moderate temperature). These two temperatures were then deceptively applied to the “lidocaine” and Vaseline cream sites throughout the conditioning and reinforcement experimental phases.

Creams were then applied to two adjacent 3x3 cm squares on the participants’ right face, arm, or leg. To enhance the believability that the “lidocaine” cream contained an active analgesic, a false label was attached to the cream bottle and green food colouring was added. The positions of the “lidocaine” and vaseline creams were counterbalanced between participants to reduce potential confounders of local sensitivity. Ten minutes following cream application, we conducted two rounds of conditioning. Participants believed they would receive eight identical moderate thermal stimuli and were instructed to report their perceived pain intensity using the VAS. Participants were also asked prior to each set of stimuli for an average expectation of the pain they would experience, which acted both to measure belief that lidocaine was working to modulate their subjective pain, and to reinforce the pain-relieving quality of the cream. During the two conditioning rounds we deceptively applied a moderate temperature to the vaseline cream site, and a low temperature to the green ‘lidocaine’ cream site.

Reinforcement and Test: At approximately the same time on the following day, sessions 2 and 3 were conducted with participants inside the MRI scanner and consisted of a reinforcement protocol (session 2) and a test protocol (session 3). The creams were applied to the same part of the face, arm or leg, in the same order and locations as session 1, but on both the left and right sides, and participants were reminded of the “lidocaine’s” pain-relieving qualities. Reinforcement was conducted by applying four noxious stimuli at the same low and moderate temperatures that were used throughout session 1 applied to the left face, arm or leg. This reinforcement protocol was conducted to ensure that participants continued to report different expectations and pain ratings between the two cream sites despite the change in day and environment (inside the MRI).

In those participants that had creams applied only to the face (n=23), a single unknown cream was also placed onto the right forearm at the commencement of the test phase, immediately prior to entering the MRI (Fig 1B). This cream was applied and described to participants to be either the control or the “lidocaine” analgesic cream, and participants could not directly observe the application of this cream nor determine its described properties. In these participants, eight noxious stimuli of moderate intensity were applied to an untreated skin site of the right forearm and pain intensity ratings collected immediately prior to commencement of the test phase as a measure of baseline pain perception.

Following reinforcement, we waited 15 minutes for residual pain and sensitivity to dissipate before beginning the test protocol. During this 15-minute period, structural brain scans were collected. Dissimilar to conditioning and reinforcement, during the test phase we applied *identical moderate temperature stimuli* to both the control Vaseline and placebo “lidocaine” cream sites on the right side of face, arm or leg (Fig 1A). We asked each participant for an average expectation of pain intensity directly prior to stimulation and instructed them to report the pain intensity experienced over the duration of the scan using the button box and the projected digital VAS. Each participant received two consecutive series of eight stimuli, with a separate functional series collected during each series of stimuli. The control cream site was always stimulated during the first series, and the “lidocaine” site was stimulated during the second series, so that we generated a “pre” and “post” condition, or, functional brain images encoding typical and placebo pain responses, respectively.

Immediately following the two face placebo scans, the thermode was reattached to the right forearm on the site where the unknown cream had been applied. Identical intensity moderate temperature stimuli as applied to the adjacent untreated site of the forearm were applied to this site, now after evoking face placebo, to determine whether the behavioural responses were isolated to the body location where response conditioning had been conducted, or whether systemic analgesic effects had been generated (Fig 1B). Mean pain intensity ratings to each stimulus applied to the two sites of the face and forearm, as well as a measure of change in pain between these two sites was calculated and both paired t-tests and linear regression analyses performed to determine whether any relationships were present in inhibition of perceived pain outside the body site conditioned to express placebo analgesia.

MRI data acquisition and preprocessing: Brain images were acquired using a whole-body Siemens MAGNETOM 7 Tesla (7T) MRI system (Siemens Healthcare, Erlangen, Germany) with a combined single-channel transmit and 32-channel receive head coil (Nova Medical, Wilmington MA, USA). Participants were positioned supine with their head in the coil and sponges supporting the head laterally to minimise movement. A T1-weighted anatomical image set covering the whole brain was collected (repetition time=5000 ms, echo time=3.1ms, raw voxel size=0.73x0.73x0.73mm, 224 sagittal slices, scan time=7mins). The two fMRI acquisitions each consisted of a series of 134 gradient echo echo-planar measurements using blood oxygen level dependant (BOLD) contrast covering the entire brain. Images were acquired in an interleaved collection pattern with a multi-band factor of four and an acceleration factor of three (repetition time=2500ms, echo time=26ms; raw voxel size=1.0x1.0x1.2mm, 124 axial slices, scan time=5:35mins).

Image preprocessing and statistical analyses were performed using SPM12 and custom software. The first five volumes of each scan were removed from the model due to excessive signal saturation from the scanner. The remaining 129 functional images were slice-time, motion corrected, and the resulting 6 directional movement parameters inspected to ensure that all fMRI scans had no greater than 1mm of linear movement or 0.5 degrees of rotation movement in any direction. Images were linearly detrended to remove global signal changes, physiological noise relating to cardiac (60-120 beats per minute $\pm$ 1 harmonic) and respiratory (8-25 breaths per minute

± 1 harmonic) frequency and harmonics modelled and removed using the DRIFTER toolbox (52), and the 6-parameter movement related signal changes were modelled and removed using a linear modelling of realignment parameters (LMRP) procedure. Each individual's fMRI image sets were then co-registered to their own T1-weighted anatomical and underwent distortion correction using the Synthetic BOLD Distortion Correction (SynBOLD DisCO) toolbox (53).

Following anatomical co-registration, the T1-weighted image was then spatially normalized to the DARTEL template in Montreal Neurological Institute (MNI) space and the parameters applied to the fMRI image sets. To focus on the brainstem specifically, the brainstem and cerebellum were isolated using the spatially unbiased infra-tentorial template (SUIT) toolbox (54), for both the fMRI and T1 image sets, and then normalised to the brainstem- and cerebellum-only SUIT template in Montreal Neurological Institute (MNI) space. During this process, both the T1 structural and functional image sets were resliced into 0.5mm isotropic voxels and these images were spatially smoothed using a 1mm FWHM Gaussian filter. A small smoothing kernel was used to allow for the accurate investigation of signal intensity changes within small brainstem nuclei (55).

Dichotomizing placebo responder and non-responder groups: For all behavioural testing, mean VAS ratings for each noxious stimulus were derived as the average pain intensity reported during plateau periods consistent with the peak of each noxious stimulus applied to either the control stimulated or 'lidocaine' stimulated series. First, the overall effect of the placebo on pain perception was first tested against all participants using a paired t-test. Participants were then grouped as either a responder or non-responder to placebo analgesia based on a bootstrapped permutation procedure (56). Briefly, mean VAS ratings to each of the 8 noxious stimuli delivered during the control stimulated series were entered to a permutation model, where 10,000 artificial sample were generated with replacement. This artificial sample was significance tested to 10,000 artificial samples generated from the VAS ratings to each 8 noxious stimuli delivered during the 'lidocaine' stimulated series. If the mean difference between the two series was significant ($p < 0.05$), with the 'lidocaine' significantly lower than the control, a participant was considered a responder. If not, they were considered a non-responder. Significant differences between groups with respect to expected changes in pain intensities immediately prior to testing were determined using paired t-tests (two-tailed, $p < 0.05$). Since participants were grouped into either responder or non-responder categories based on their perceived pain intensities during the fMRI scans (session 3), we did not assess significant differences between groups for the perceived pain intensity changes. A single factor ANOVA ($p < 0.05$) was also used to determine if there were differences in the temperature applied or pain intensity ratings reported between responder and non-responder groups during the control stimulated series to ensure any reported placebo effects did not relate to baseline thermal sensitivity. In addition, we calculated each participants analgesic ability by calculating the difference in mean pain intensity ratings during the control compared with the placebo scan (analgesic ability = control scan average pain intensity minus placebo scan average pain intensity). A greater (more positive) analgesic ability value represents a greater pain reduction during the placebo compared with control scan.

fMRI statistical analysis: To determine significant changes in signal intensity during each noxious thermal period, a repeating boxcar model convolved with a canonical hemodynamic

response function was applied to each of the fMRI series. The contrast images generated for each functional image series were then used in group analyses. Following the creation of a PAG and RVM mask drawn on a mean brainstem fMRI image set calculated from one image from each of the 93 participants, we conducted two separate analyses to determine PAG and RVM signal changes during placebo:

responders versus non-responders: significant differences between PAG and RVM signal intensity changes of responder and non-responder groups were determined by subtracting the control scan contrast map from the placebo scan contrast map in each individual. The signal intensity change difference maps (placebo lidocaine – control scan) were then compared between responder and non-responder groups using random effects, two-sample voxel-by-voxel analyses. The greatest significantly different PAG or RVM cluster in the responder<non-responder contrast was determined using a minimum threshold of $p < 0.005$ (cluster extent threshold of five contiguous voxels) and overlaid onto a mean brainstem T1-weighted anatomical image set located in the same exact location as the mean brainstem fMRI image set. The signal change differences (β values) were extracted from each significant cluster and the mean $\pm$ SEM values plotted.

linear correlation with analgesia: significant relationships between PAG and RVM signal change differences and analgesic responses were determined by placing the signal intensity change difference maps (placebo-control scan) compared with analgesic ability values into a random effects, one-sample, voxel-by-voxel analyses. The greatest negative or positive correlation PAG and RVM cluster was determined using a minimum threshold of $p < 0.005$ (cluster extent threshold of five contiguous voxels) and overlaid onto a mean brainstem T1-weighted anatomical image set. The signal change differences (β values) were extracted from each significant cluster and plotted against analgesic abilities and rho and p values (Pearsons correlation coefficient) calculated. **To evaluate regional specificity in each PAG and RVM cluster demonstrating linear relationships with analgesic ability, significant clusters of each independent body site were saved as region-of-interest masks, and signal intensity change values extracted from the alternate two body sites. Linear regression was conducted with rho and p-values determined for all three body sites in each PAG and RVM cluster, with each pair of regression slopes between two body sites compared to determine whether linear relationships between activation change in either the PAG or RVM, and analgesic ability on the face, arm, or leg, were significantly different.**

In addition, since we have previously shown that the columnar organization of the PAG is more accurately determined by assessing the location of signal intensity changes at an individual participant level, we performed a third analysis:

individual placebo analgesia PAG peaks in responders: we determined the location of the lowest nadir of the placebo – control contrast maps in the PAG in each responder for the face, arm and leg. To determine the location of this peak with respect to PAG columns, masks encompassing the dorsomedial (dmPAG), dorsolateral (dlPAG), lateral (lPAG) and ventrolateral PAG (vlPAG) columns as defined by Bandler and Keay (3) were created at 1mm intervals throughout its rostrocaudal extent (Z co-ordinate -3 to -13) (Supplementary Fig 3). We then recorded and plotted the mean $\pm$ SEM MNI co-ordinates onto a mean brainstem T1-weighted anatomical image set. As

the focus of this study was on investigating rostro-caudal differences within the PAG, a one-way ANOVA was performed to compare the effect of body site (face, arm and leg) on the mean ipsilateral Z co-ordinate. The Levene test was first used to ensure the assumption of homogeneity of variances was met as well as the Shapiro-Wilk test for normality across the Z-coordinates for the face, arm and leg. Post-hoc analyses were then performed using Tukey's honest significance difference (HSD) test to determine whether there was a significant difference in the mean Z-coordinates between the face, arm and leg.

In each of the responder versus non responder and linear regression analyses, small volume correction (SVC) was applied to resulting clusters at a corrected family wise error p-value threshold of $p < 0.05$. This step was performed by creating two millimetre spherical region-of-interest masks consistent with the somatotopic arrangement of the PAG to noxious heat previously identified in (6). These co-ordinates were extracted and translated along the x-axis for application of SVC to ipsilateral PAG clusters.

Supplementary Information:

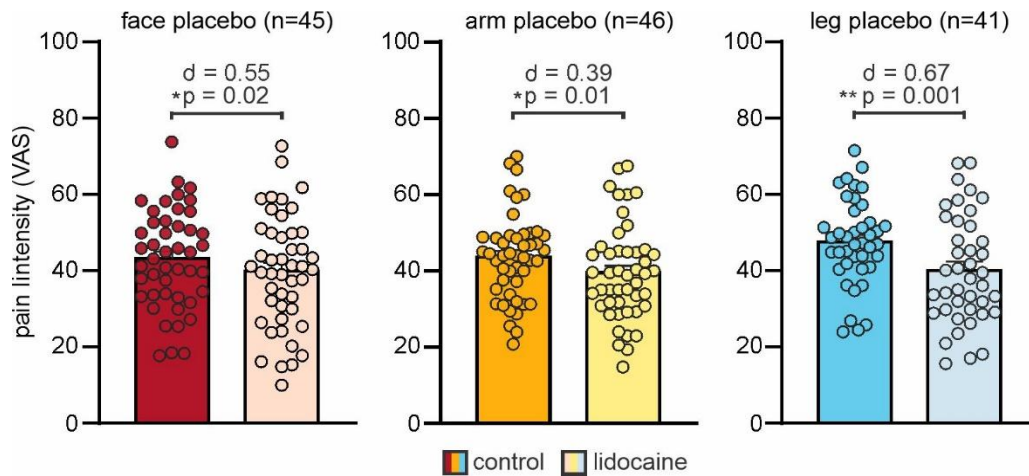


Fig. S1.

Main effect pain intensity ratings during test phase across entire cohort in face, arm, and leg placebo. At each body site we observed a significant behavioural effect of placebo analgesia, however, this result comprises both significant placebo analgesia responders (those who demonstrated a reduction in pain during identical stimulus intensity application to both control and placebo lidocaine cream sites) and non-responders (those where pain either did not change or increased).

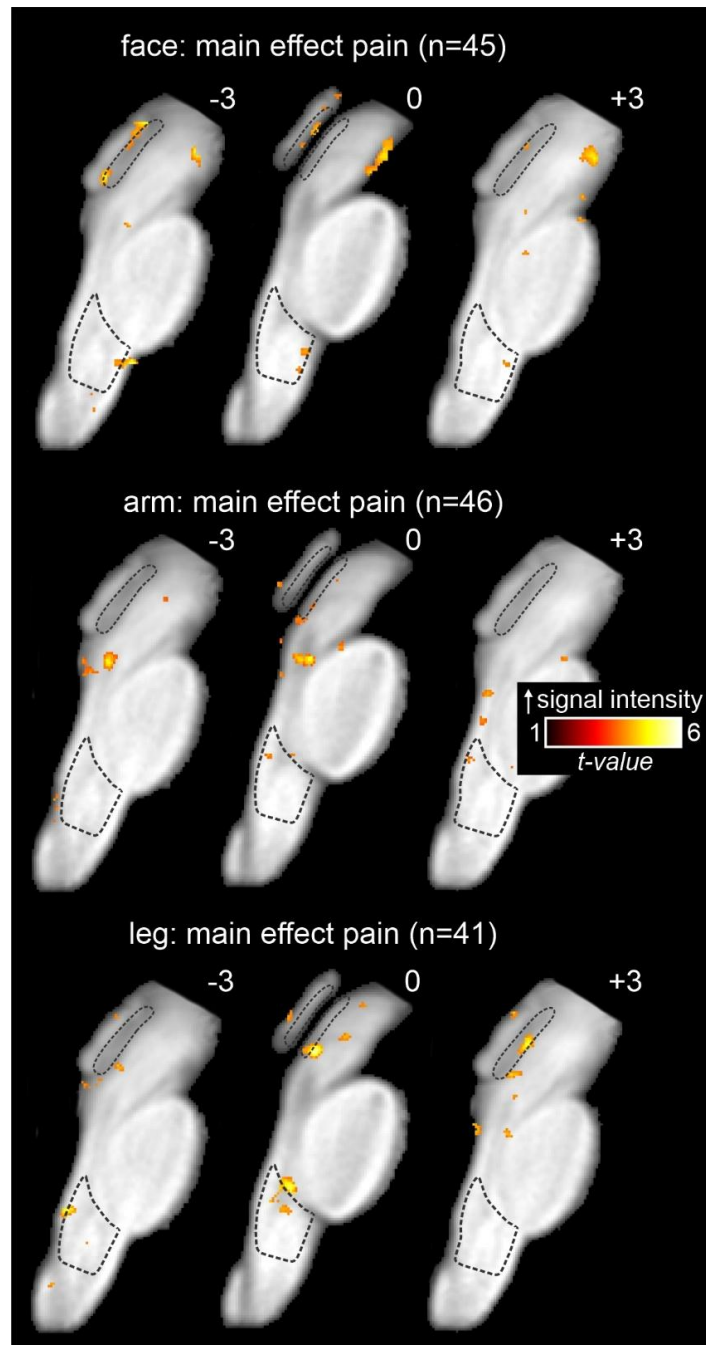


Fig. S2.

Brainstem main effect of pain activation contrast images. Analysis of brainstem-wide activation change at a cluster forming threshold of $p < 0.001$ uncorrected for multiple comparisons during noxious stimulation of the face (upper row), arm (middle row), and leg (lower row) in all participants prior to evoking placebo analgesia. Sagittal plane X co-ordinates in Montreal Neurological Institute space are provided in the upper left corner of each image. **Rostral and caudal dotted outlines mark the anatomical boundaries of the midbrain periaqueductal gray matter, and rostral ventromedial medulla, respectively.**

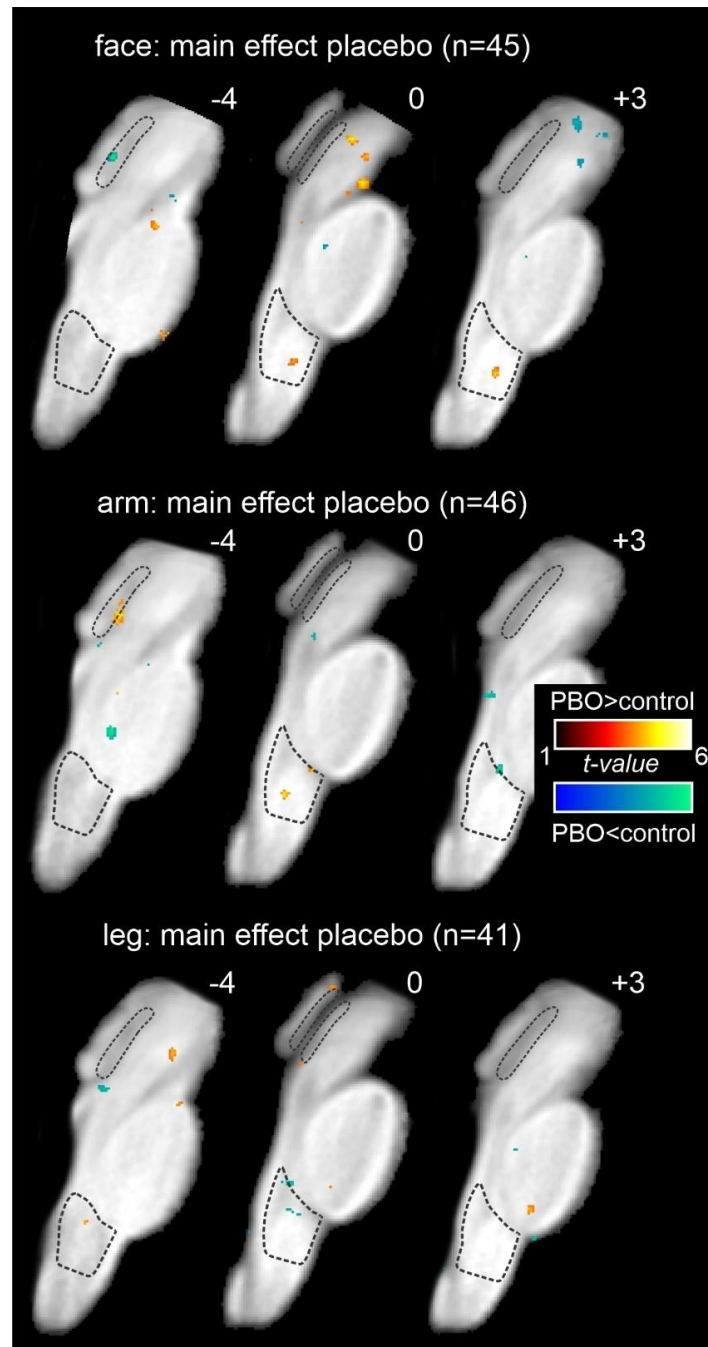


Fig. S3.

Brainstem main effect of placebo activation contrast images. Analysis of brainstem-wide activation change at a cluster forming threshold of $p < 0.001$ uncorrected for multiple comparisons during noxious stimulation of the placebo relative to control cream sites of the face (upper row), arm (middle row), and leg (lower row) in all participants. Sagittal plane X co-ordinates in Montreal Neurological Institute space are provided in the upper left corner of each image. Rostral and caudal dotted outlines mark the anatomical boundaries of the midbrain periaqueductal gray matter, and rostral ventromedial medulla, respectively.

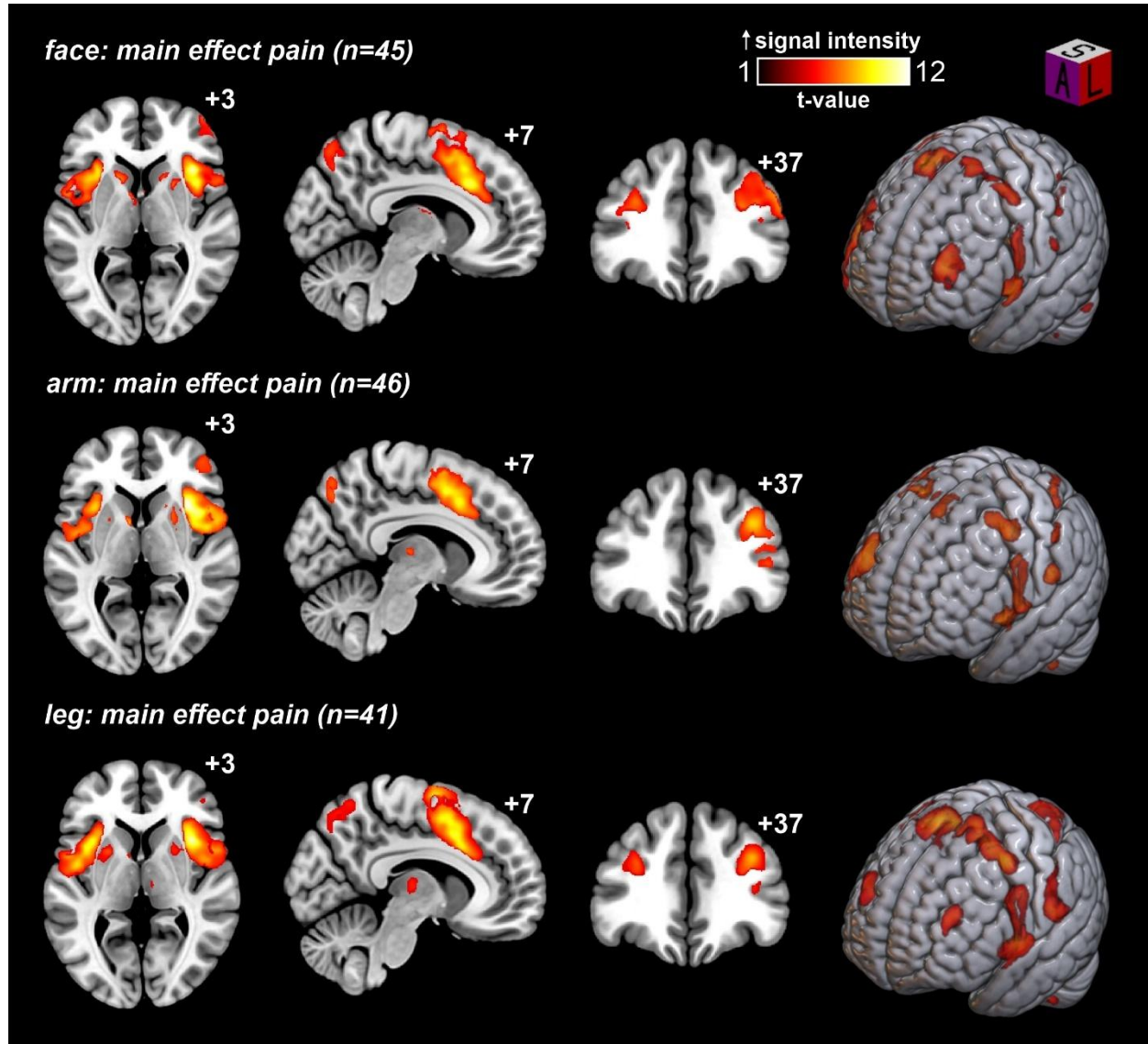


Fig. S4.

Cortical main effect of pain activation contrast images. Analysis of brainstem-wide activation change at a family-wise error corrected cluster forming threshold of $p < 0.05$ during noxious stimulation of the face (upper row), arm (middle row), and leg (lower row) in all participants prior to evoking placebo analgesia. Co-ordinates in Montreal Neurological Institute space are provided in the upper right corner of each image. Left facing renders on the far-right panel demonstrate the extent and intensity of cortical activation.

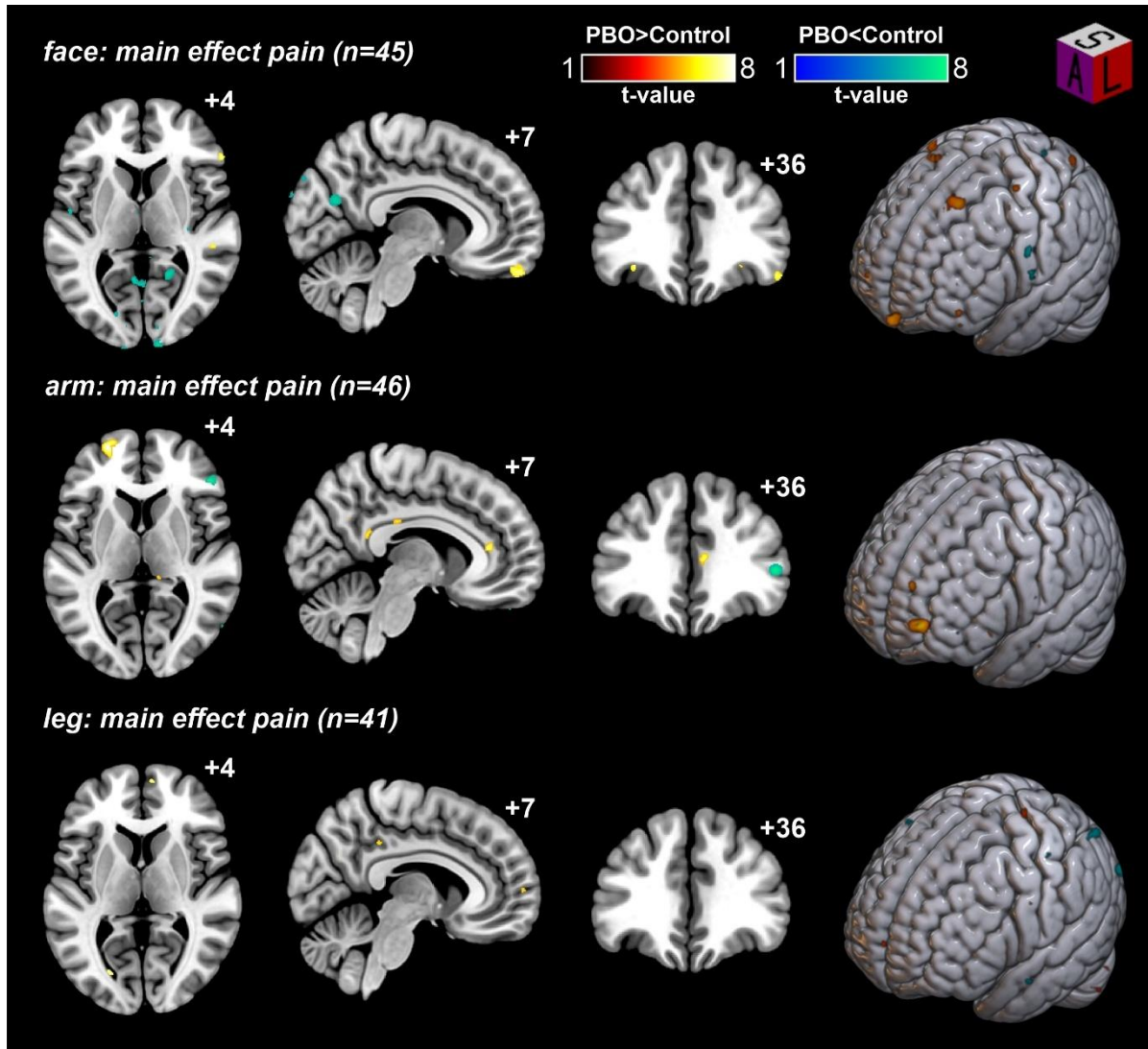


Fig. S5.

Cortical main effect of placebo activation contrast images. Analysis of brainstem-wide activation change at a cluster forming threshold of $p < 0.001$ uncorrected for multiple comparisons during noxious stimulation of the placebo relative to control cream sites of the face (upper row), arm (middle row), and leg (lower row) in all participants. Co-ordinates in Montreal Neurological Institute space are provided in the upper right corner of each image. Left facing renders on the far-right panel demonstrate the extent and intensity of cortical activation.

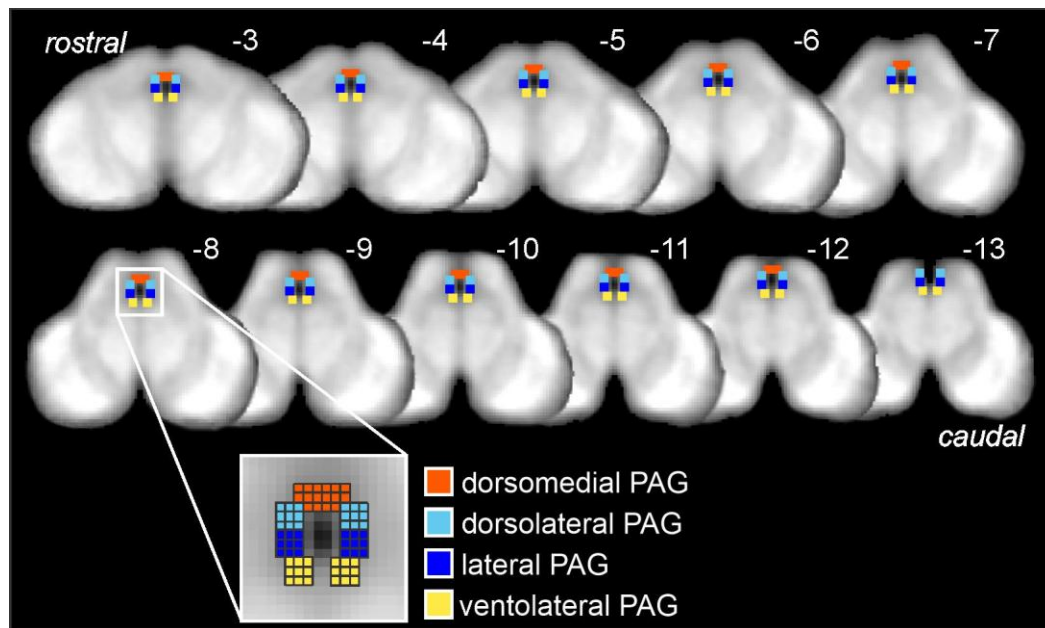


Fig. S6.

Midbrain periaqueductal gray matter mask images. To resolve which column of the midbrain periaqueductal gray matter individual nadir signal change points during placebo analgesia expression of the face, arm, or leg occupied, masked areas of each of the four longitudinal PAG columns were generated, and co-ordinates entered in viewing software.

face		arm		leg	
male (n=23)	female (n=22)	male (n=24)	female (n=22)	male (n=20)	female (n=21)
<i>age (mean±SEM years)</i>					
27.7±1.2	28.1±2.5	24.8±0.8	23.7±0.8	24.9±0.7	23.7±0.8
<i>response distribution (responder : non-responder)</i>					
12:11	10:12	15:9	8:14	11:9	11:10
<i>thermode temperature (mean±SEM)</i>					
46.2±0.2	46.2±0.2	46.8±0.2	46.7±0.2	46.4±0.1	46.3±0.1

Table S1.

5 Between sex differences in mean age, response distribution, and elected moderate thermode intensity in each body site subject to response conditioning and evaluated for placebo analgesia expression.

	MNI co-ordinate		
	X	Y	Z
<i>PAG individual maxima (mean co-ordinate±SEM)</i>			
face (n=45)	1.65±0.10	-30.67±0.38	-6.17±0.41
arm (n=46)	1.76±0.09	-32.00±0.40	-7.93±0.37
leg (n=41)	1.80±0.12	-32.28±0.36	-8.08±0.45

Table S2.

Whole sample individual peak X, Y and Z Montreal Neurological Institute (MNI) co-ordinate of the ipsilateral midbrain periaqueductal gray matter (PAG) during placebo analgesia paradigm applied to the face, arm and leg.

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Supplementary Materials

Figs. S1 to S6

Tables S1 to S2